

Project Name: QQQ Investment Strategy Analysis

Objective:

To develop a Python-based tool that analyzes historical data of QQQ to identify optimal investment schedules and amounts, aiming to maximize returns for investors.

Key Considerations:

1. **Historical Data Acquisition:** Fetch past performance of QQQ to identify patterns or trends.
2. **Investment Strategy Simulation:** Implement two simple investment strategies - The Dollar-Cost Averaging (DCA) strategy and momentum investing strategy and simulate their performance over historical periods.
3. **Return calculation:** Calculate annualized return, total return for both investment strategy in the range of [01-01-2014, 12-31-2023]
4. **Risk Assessment (optional):** Evaluate the risk associated with each strategy using metrics like maximum drawdown, volatility, and Sharpe ratio.

Tools and Libraries:

- Language: Python 3.7+
- Data Analysis lib: pandas, numpy
- Data Visualization: matplotlib, seaborn
- Financial Data API: yfinance (to fetch historical data of QQQ) or Stooq

Steps:

1. **Setup Environment:**
 - a. Ensure Python 3.7+ is installed.
 - b. Install required libraries: **pandas, numpy, matplotlib, seaborn, yfinance**
2. **Data Acquisition:**
 - a. Use **yfinance** to download historical data of QQQ.
 - b. if can't connect to yfinance, use pandas (stooq)

```

1  import yfinance as yf
2
3  #1
4  qqq_data = yf.download('QQQ', start=start_date, end=end_date)
5
6  #2
7  qqq_data = web.DataReader('QQQ', 'stooq', start_date, end_date)

```

3. Data Preprocessing:

- a. Clean the data for any missing values or anomalies.
- b. Calculate additional indicators that may be useful (e.g., moving averages).

4. Strategy Implementation:

- a. Implement multiple investment strategies for comparison.
- b. Each strategy should define when and how much to invest in QQQ.
- c. DCA: The Dollar-Cost Averaging (DCA) strategy is an investment technique designed to reduce the impact of volatility on scheduled purchases of financial assets such as stocks, in this case, use monthly investment for QQQ with \$500 every time.
- d. momentum investing strategy: Momentum investing is a strategy that involves buying securities that have had high returns over a recent period and selling those that have had poor returns over the same period. The core idea behind momentum investing is "what goes up must continue to go up," at least in the short term.

```

1  # DCA, never sell the stock
2  shares_bought = monthly_investment / monthly_data
3
4  # momentum investing strategy
5  # Calculate 6-month momentum, skipping the most recent month
6  momentum = monthly_returns.rolling(window=6).apply(lambda x:
7  x[:-1].sum())
8  # Determine positions: 1 for invest in QQQ, 0 for cash
9  positions = (momentum > 0).astype(int)

```

5. Simulation:

- a. Simulate each investment strategy over the historical data.
- b. Calculate performance metrics for each strategy.

6. Return calculation:

- a. Assess the risk associated with each optimized strategy.

```
1  fun backtest_dca(ticker, start_date, end_date,  
    monthly_investment):  
2      return annualized_return, total_return  
3  
4  fun momentum_strategy(ticker, start_date, end_date,  
    monthly_investment):  
5      return annualized_return, total_return  
6  
7  # expected return for the last 10 years would be around 2.7 times  
    of the total invested for both strategies
```

7. Risk Assessment:

- a. Assess the risk associated with each optimized strategy.
- b. check the public documentation to calculate maximum drawdown, volatility, and Sharpe ratio.
- c. reference:

1. Total Return

Total return measures the overall return of an investment over a specific period, including both capital gains and dividends.

Formula:

$$\text{Total Return} = \frac{\text{Ending Value} - \text{Beginning Value} + \text{Dividends Received}}{\text{Beginning Value}} \times 100\%$$

2. Annualized Return

Annualized return, or compound annual growth rate (CAGR), measures the mean annual growth rate of an investment over a specified time period longer than one year.

Formula:

$$\text{Annualized Return} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{n}} - 1 \times 100\%$$

Where n is the number of years.

3. Volatility (Standard Deviation)

Volatility measures the dispersion of returns for a given security or market index. It's often used as a measure of risk.

Formula:

$$\text{Volatility} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (R_i - \bar{R})^2}$$

Where R_i is the return in period i , $\bar{R}$ is the average return, and N is the number of periods.

4. Sharpe Ratio

The Sharpe ratio measures the performance of an investment compared to a risk-free asset, after adjusting for its risk.

Formula:

$$\text{Sharpe Ratio} = \frac{\text{Return of Portfolio} - \text{Risk-Free Rate}}{\text{Volatility of Portfolio}}$$

